Modeling and Analysis of a Pendulum-Cart Coupled System

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Nomenclature

c = cart damping coefficient $[\frac{n \cdot s}{m}]$

b = pendulum damping coefficient $[\frac{n \cdot s}{rad}]$

g = gravitational constant $[\frac{m}{s^2}]$

k = spring constant $[\frac{n}{m}]$

M_1 = mass of cart [Kg]

M_2 = mass of pendulum [Kg]

P = Period [s]

R = pendulum arm [m]

x = linear position [m]

$\dot{x}$ = linear velocity $[\frac{m}{s}]$

$\ddot{x}$ = linear acceleration $[\frac{m}{s^2}]$

δ = log decrement parameter

Δ = uncertainty in measurement

ζ = damping coefficient

θ = angular position [rad]

$\dot{\theta}$ = angular velocity $[\frac{rad}{s}]$

$\ddot{\theta}$ = angular acceleration $[\frac{rad}{s^2}]$

ω_d = dampened natural frequency

ω_n = undamped natural frequency

I. Introduction

A mass damper system consisting of a pendulum on a swinging cart grounded via a spring can be analyzed and modelled as a coupled dynamics problem which is subject to non-linear motion. The internal coupling of masses presents possible optimization via vibration reduction, which will be explored via the generation of a computational and mathematical model. These numerical solutions will serve as simulations of the free response output of the system. The integrity of these models will be verified against the results from the real dynamics of the model.

Pendulums can be utilized to resist steady state and low seismic motion such as wind bellowing against a building. As passive tuned and active dampeners, pendulums are incredibly useful tools in vibrational analysis. However, it is critical when modelling any system susceptible to disturbances that the dynamics are fully understood. This ensures a wide range of input conditions can be explored and ensures a system is safe prior to deployment. The fundamental methodology explored in this lab follows the same principles which would be faced in the real-world deployment of dampeners and serves as an example to be extrapolated.

II. Experimental Methods

The experiment was set up with two components. The computer running the .vi interface which was able to monitor and record the various sensor input, and the mass pendulum system (Fig. 1).

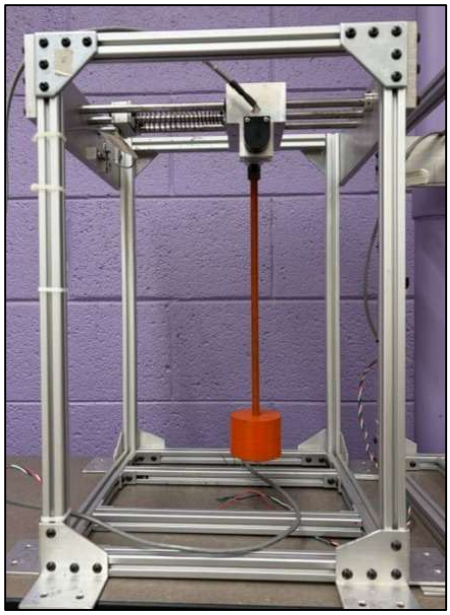


Fig. 1 Mass pendulum system experimental setup.

There are two sensors in the system. The first is the angular encoder. Angular encoders work by assigning every angle a visually distinct marker. These markers are read by a recording device, which outputs the corresponding angle. This is paired alongside a strain gauge. The strain gauge is connected in series to the spring. This location means that any force the spring outputs will cause slight deformation, and therefore, strain in the metal connection. This slight deformation is measured by the gauge, and recorded as voltage. This functions as an accurate measure of distance due to the nature of a spring. The spring, while unstressed, will rest at a neutral position. At this position, there is no force exerted on the strain gauge, and therefore this is the equilibrium position. Any movement of the cart will transmit force to the strain gauge following Hooke's law. Hooke's law is below.

$$F = -kx \tag{1}$$

The linear relationship of Hooke's law leads to linear output from the strain gauge. The strain gauge was implemented here as it is a more accurate sensor when compared to the alternatives such as ultrasonic.

The cart slides on two bars and a spring. The two bars behave as dampeners through friction, and are represented as such in the free body diagram. There is a ruler fastened to one of the rails on top, which will be used to calculate the spring constant. The entire system is fastened to the table to provide a static foundation for the system.

The spring used in this experiment has an unknown k value, which will need to be found experimentally. The spring constant is a system parameter, and was calculated using a ruler mounted to the top of the assembly, and a force gauge. The cart was pulled along the rails until the force gauge read 3N, and 4N. At both of these forces, distance measurements were taken. This step was repeated twice. This change in distance was divided by the change in force. The equation used to calculate the spring constant is Hooke's law rearranged.

$$k = \frac{\Delta x}{\Delta F} \tag{2}$$

With the spring constant calculated, there were a few more measurements to collect. The pendulum (M_2) was placed on the scale and measured, and the moment arm of the pendulum was measured using a tape measure.

With the necessary physical system parameters identified, the data collection could begin. The first experiment involved measuring the entire system response. The pendulum was swung, and the response moved the pendulum back and forth. The force of the pendulum caused the cart to slide back and forth along the track. This response was collected through the two sensors discussed earlier. Through the strain gauge and angular encoder, the response of the complete system will be plotted.

The next experiment involved fastening the cart using shaft collars in order to fix its position. This removes the forces from the spring and dampers with the purpose of isolating the pendulum. The pendulum was swung, and data was collected on its angular position versus time. Without the cart, the pendulum loses little angular momentum, and ample time was allowed for swinging to ensure many data points were collected at the ends of each swing.

Finally, the pendulum was removed from the cart, and the cart was pushed. The force of the spring, k, and the damping force, c, still remained. This brought the cart to a quick stop after a few oscillations around its equilibrium point. The purpose of testing the car and pendulum separately was to isolate their respective damping properties before beginning the mathematical analysis.

III. Mathematical Analysis

A. System of Equations

The response of a dynamic system like the one in question, is defined by the system parameters. This system involves a spring, two dampening forces, and the mass of both the cart and the pendulum. These can be seen labeled in Fig. 2.

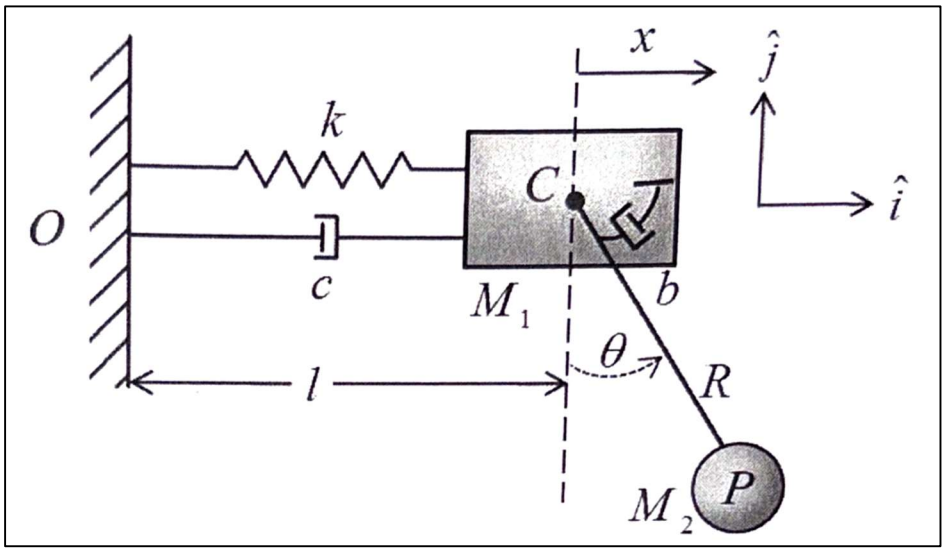


Fig. 2 Free body diagram of mass pendulum system.

These are the parameters that will be solved through mathematical analysis, although k, M_2, and R were found experimentally. This leaves M_1, c, and b to be found through mathematical analysis. There are other values that are important to a dynamic system other than those labeled above. These values will prove necessary to calculate all of

the system parameters. These include the damping ratio, ζ which represents how quickly the system returns to equilibrium. The two natural frequencies, ω_n, and ω_d, which represent the natural frequency, and the damped natural frequency respectively. To solve the system parameters, the equations of motion must first be generated. This was done using the Lagrangian method. The complete derivation can be found in the appendix. These derivations result in the nonlinear equations of motion seen in equations (3) and (4).

$$(M_1 + M_2)\ddot{x} + c\dot{x} + kx = M_2 R\dot{\theta}^2 sin\theta - M_2 R\ddot{\theta} cos\theta \tag{3}$$

$$M_2 R\ddot{\theta} = -M_2 \ddot{x} cos\theta - \frac{b}{R} - M_2 g sin\theta \tag{4}$$

This pair of equations represent the system response of both the cart and the pendulum. These two equations will be simulated in Simulink, with their responses plotted. These plots will be compared to the plots from the experimental analysis. In addition to this, the equations of motion will be linearized, and simulated again in Simulink. To linearize equations (3) and (4), the small angle approximation will be used.

The small angle approximation works by replacing the non-linear terms in an equation with linear representations. These linear representations are found through a first order Taylor series approximation. For the small angle approximation, the linearization is done around zero, because this is the resting point for the pendulum. The nature of the small angle approximation being a linear approximation of a non-linear curve leads to inaccuracies. This means that a larger angle will lead to more error in the approximation. The small angle approximation is derived in the appendix. The linearized equations can be seen below, with derivations in the appendix.

$$(M_1 + M_2)\ddot{x} + c\dot{x} + kx = M_2 R\dot{\theta}^2 \theta - M_2 R\ddot{\theta} \tag{5}$$

$$M_2 R\ddot{\theta} = -M_2 \ddot{x} - \frac{b}{R} - M_2 g\theta \tag{6}$$

B. System Parameters

As covered earlier, there are only a few system parameters remaining for the calculations. M_1, c, and b all need to be found. M_1 will be calculated first, but will require multiple other parameters to be found. The calculations for M_1were completed using the data with the pendulum removed. This was for isolating the cart, and ensuring the parameters are for the cart exclusively, and not for the system. The first calculation to be completed was the period. The period was calculated from the first two local maximums of the response. The data points used for the period calculations are below.

$$(x_1, y_1), (x_2, y_2) = (1.75815, 2.703194), (2.318526, 2.398450) \quad (7)$$

Using the formula for period listed below, the period can be calculated.

$$P = x_2 - x_1 \quad (8)$$

The period will be used to calculate the dampened natural frequency (ω_d) below.

$$\omega_n = \frac{2\pi}{P} \quad (9)$$

The log decrement method will be used for calculating the damping ratio (ζ). This involves finding the ratio of the subsequent peaks and using the natural logarithm to define a variable representing this decay (δ). The cart response is plotted below in FIG. 3 with both points marked, along with a visual for the log decrement method.

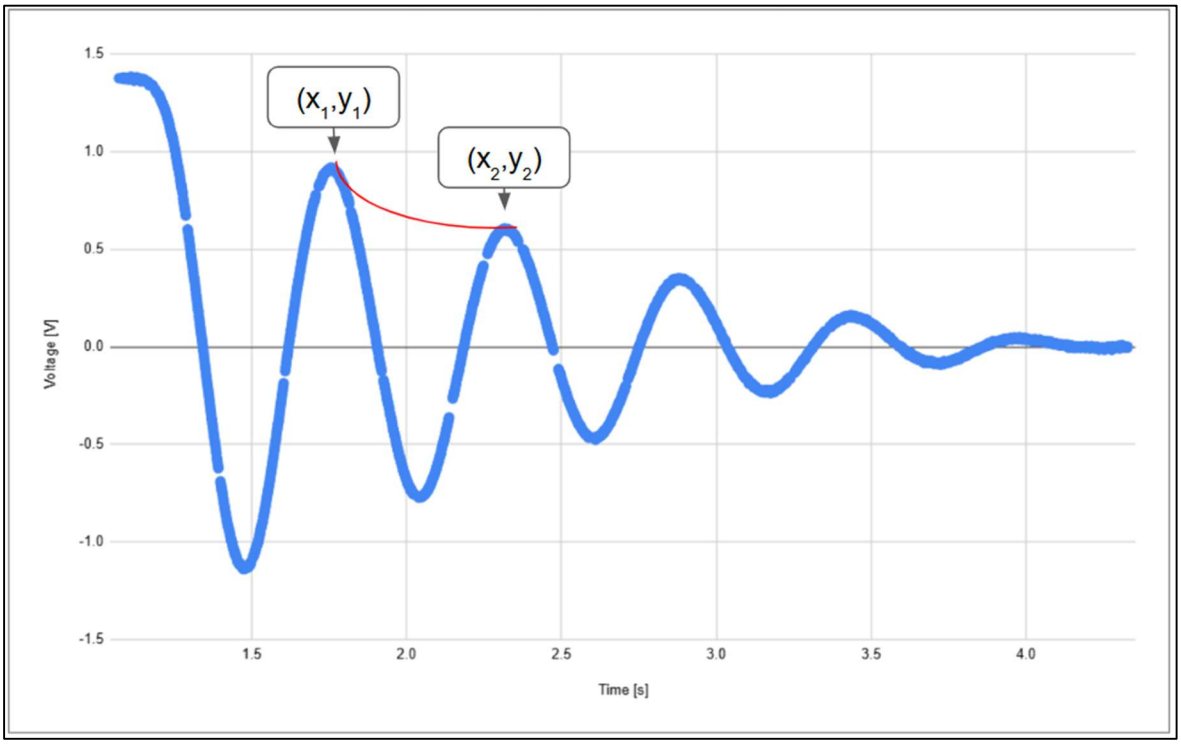


Fig. 3 Cart response with log decrement method.

The general form of the log decrement method equation is listed below where n is the number of peaks between the datapoints.

$$\delta = \frac{1}{n} \ln\left(\frac{x_1}{x_{n+1}}\right) \quad (10)$$

Delta is useful due to its relationship with the damping ratio. This relationship is listed below, along with the simplification to solve for the damping ratio.

$$\delta = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}} \Rightarrow \zeta = \frac{\delta}{\sqrt{4\pi^2+\delta^2}} \tag{11}$$

Using this relationship to solve for the damping ratio, the damping ratio and the dampened natural frequency can be used to solve for the undampened natural frequency. This relationship is shown below.

$$\omega_d = \omega_n\sqrt{1-\zeta^2} \tag{12}$$

This undampened natural frequency is a requirement for the calculation of the mass of the cart (M_1). The equation for the cart's mass is below, and uses the spring constant found experimentally.

$$M_1 = \frac{k}{\omega_n^2} \tag{13}$$

With the M_1 calculated, all of the parameters required to calculate the damping coefficient c have been found.

$$c = 2\zeta\sqrt{kM_1} \tag{14}$$

This concludes all of the required parameters for the cart as a standalone system. The next part of the analysis will be using the shaft collars to isolate the pendulum as a standalone system.

The only parameter remaining is the damping coefficient b. Solving for this parameter will be reusing all of the previous equations from a different set of starting points. Instead of using the cart system, this will use the pendulum encoder data. The datapoints from the pendulum response used to calculate c are below.

$$(x_1, y_1), (x_2, y_2) = (2.901457, 23.906238), (17.160942, 23.378894) \tag{15}$$

Most equations will remain the same, equation (8) will be used to solve for the period, with equation (9) solving for the undampened natural frequency. The log decrement method will be different due to using amplitudes that are (12) apart. In this case, n = 12. With delta, solve for the damping coefficient in equation (11), and the dampened natural frequency in equation (12). The equation for the damping coefficient b is below.

$$b = 2\delta\zeta\omega_n R^2 \tag{16}$$

With b solved, all of the system parameters have been identified and solved.

C. Uncertainty

The uncertainty of the system parameters was calculated using the method of propagating uncertainty. This method ensures that the uncertainty of a measurement will be calculated for the values calculated from this measurement, but also any further calculations using those values. The general form of the uncertainty propagation equation is below.

$$\delta f = \sqrt{(\frac{\delta f}{\delta x})^2 + \cdots} \tag{17}$$

To identify the necessary application of uncertainty propagation, uncertainty propagation diagrams were made, and can be found as FIG. A1 and A2. The specific uncertainty equations, alongside the necessary values for calculation are in the appendix. The uncertainty values will be assessed in the results.

With the linear, and nonlinear equations of motion derived, the software Simulink will be used. The purpose of this simulation will be to determine the effectiveness of linearization, as well as compare both results to the experimental results.

IV. Simulation

To analyze the system's response, both linear and nonlinear models were made in MATLAB Simulink. The nonlinear model, shown in Figure 4, attempts to simulate the system's equations of motion, which include its translational and rotational behavior. The top half of the model represents the first equation and is denoted by “x”. The bottom half of the model includes the second equation and is denoted by “θ.” The initial condition of each equation comes from their respective second integrator signal, and follows the path of the model through various blocks (gain, product, etc.). Eventually, the signal loops back around before finishing at a Scope block, which was used to visualize the behavior of the system, as seen in Figures 5 and 6.

The linear model is a much simpler version of the system because it removes most of the operations/blocks that were in the non-linear Simulink model. The interconnections in the linear model follow the same signal path of the nonlinear version. This allows for the direct comparison between ideal linear and the more complex nonlinear models and their behaviors.

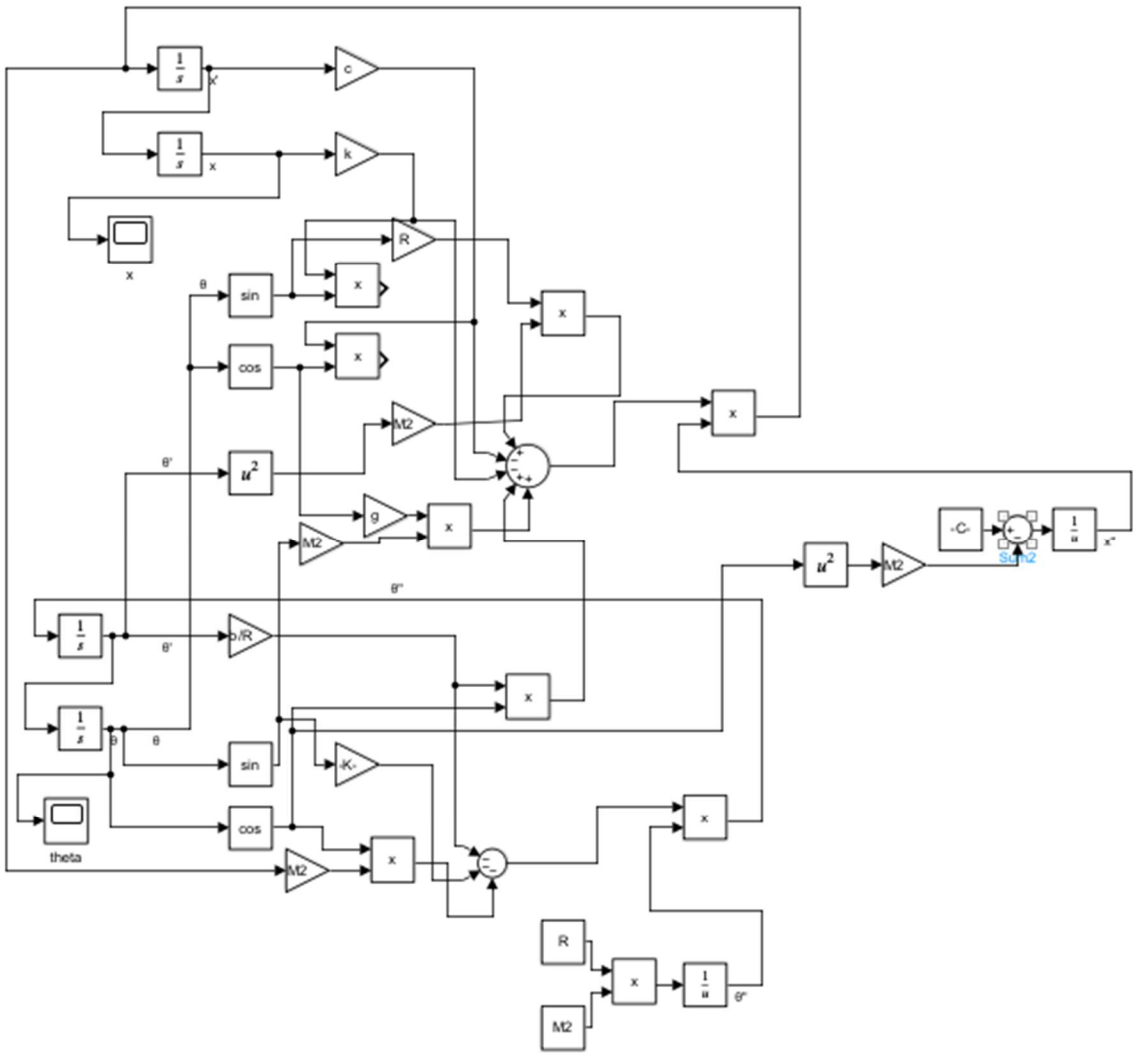


Fig. 4. Non-linearized MATLAB Simulink model of Mass-Pendulum System

By modeling the system in MATLAB's Simulink extension using its equations of motion, the response of the system at given an input can be determined. In Figure X shown below, the y-axis represents the position of the cart in meters, and the x-axis represents the time in seconds. The position of the cart from zero seconds (when cart is released from starting position) to ten seconds is shown after an arbitrary starting position of 0.87 meters and initial pendulum angle of 1 radian were provided as the inputs. This value was chosen to appropriately represent the system and its behavior. The plot shows two essential attributes of the mass-pendulum system: the damping, oscillatory motion and the erratic movement of the pendulum caused by the cart (or vice versa) as the amplitude decreases. As time goes on, the magnitudes of the angular velocity of the pendulum and linear velocity of the cart decrease as the system approaches its equilibrium position of theta = 0. As these speeds decrease, the swaying of the pendulum has a greater

effect on the movement of the cart. This can be seen as the position plot becomes less smooth as time increases. The results of these simulations can be seen below in Fig. 4 and 5.

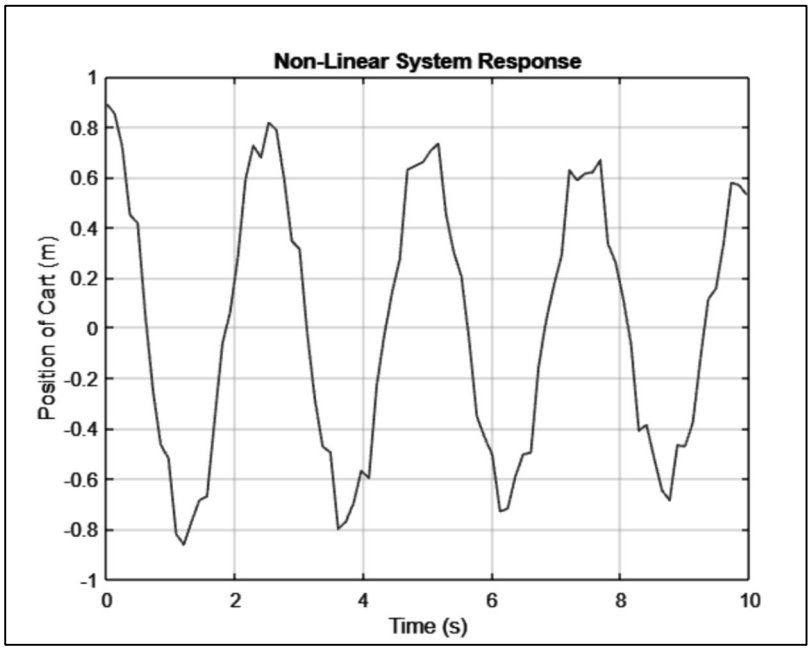


Fig. 5. Position vs Time Plot of Non-linearized Mass-pendulum System in Simulink

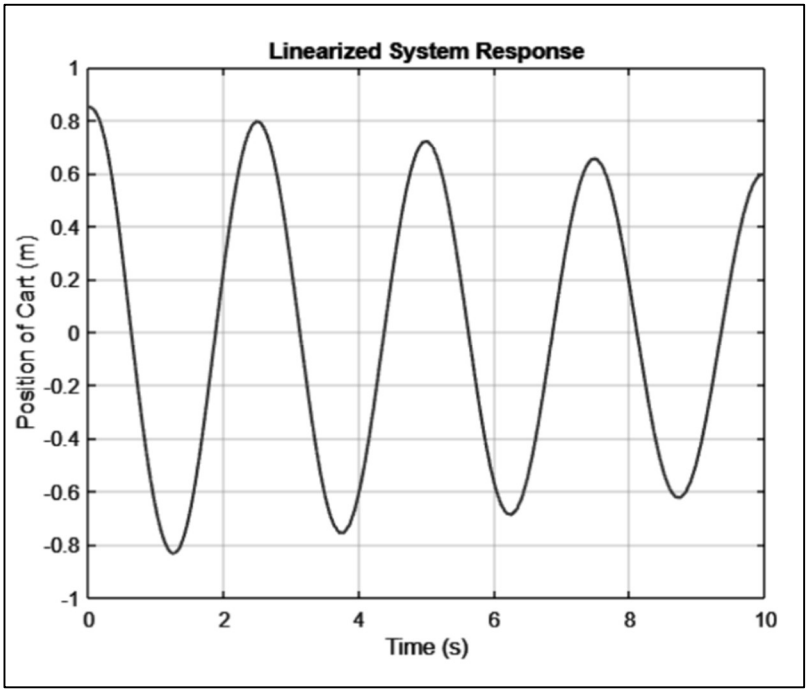


Fig. 6. Position vs Time Plot of Linearized Mass-pendulum System in Simulink

V. Results

A. System Parameters

This experiment and subsequent data analysis yielded many results. These will begin by showing the parameter values derived and calculated in the analytical methods. The results for the cart are below in Table 1.

Table 1 Cart parameter values

P [s]	ω_d	k [n/m]	δ	ζ	ω_n	c[ns/rad]	M_1[kg]
0.560376	11.212445	107.426	0.119613	0.0190333	11.2144796	0.365	0.854

The results for the pendulum are similar and will be discussed together with the cart results. The results for the pendulum can be seen below in Table 2.

Table 2 Pendulum parameter values

P [s]	ω_d	R [m]	δ	ζ	ω_n	b[ns/rad]	M_2[kg]
1.188290417	5.287584	0.370	0.0018588	0.0002958	5.2875842	0.00056963	1.330

These parameters define the nonlinear mass pendulum system. These systems behave differently when isolated from one another. This can be seen through the damping ratio. The damping ratio of the cart is many orders of magnitude larger than the pendulum. This can also be seen from the data in Fig 4 and 7.

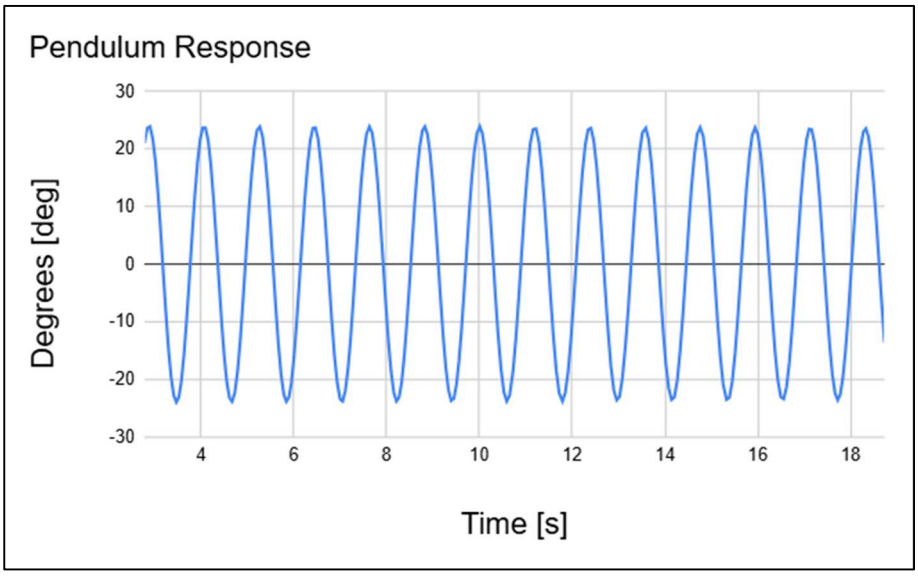


Fig. 7. Isolated pendulum response

The pendulum response hardly decays over the 18 second data collection period, and therefore is rewarded with a very small damping coefficient. This transfers through to having a small damping coefficient due to there not being much dampening at all. This can be contrasted with the comparatively high damping coefficient for the cart. This can be traced back to a high damping ratio as the cause. This implies that the cart experiences lots of friction when sliding on its rails. These values mirror what would be expected from them, and this can be reinforced through uncertainty propagation.

B. Uncertainty

The uncertainty propagation method was briefly covered in the mathematical analysis section, and the results will be shown here. All derived equations used to calculate these values can be found in the appendix along with the measurement instruments, the recorded values, and the resolutions. Table 3 contains the uncertainties for the cart parameters.

Table 3 Cart parameter uncertainty values

k[n/m]	δ	ζ	ω_n	c[ns/rad]	M_1[kg]
± 6.671	± 0.00005	± 0.000008	± 0.00000016	± 0.005	± 0.045

The uncertainty is notable for k due to its high value. This is due to the large resolution of the force gauge. The force gauge was difficult to use to obtain a proper measurement. Even with the difficulty of fighting the gauge, the resolution was only 0.2N. This was orders of magnitude greater than for example, the angular encoder which had a resolution of 0.001 Rad. The remaining values for the pendulum are below in Table 4.

Table 4 Pendulum parameter uncertainty values

R[m]	δ	ζ	ω_n	b[ns/rad]	M_2[kg]
± 0.001	± 0.00000011	± 0.000000017	± 0.00000000001	± 0.0000000018	± 0.001

The uncertainty values for the pendulum are very small. Theses are due to the accuracy of the sensor, and the incredibly slow decay from the small dampening coefficient. The remaining measurements come from more accurate instruments such as a tape measure, and a digital scale.

Uncertainty in the experimental system parameters also contributed to the differences between simulation and the experiment. The spring contact was obtained through physical measured force and displacement. The uncertainty in this method of measurement propagated into the calculated value of the spring constant, k. Since the cart's natural frequency depends on the stiffness and mass, any of the errors in the spring constant will change the oscillation frequency. This can cause the simulated and experimental peaks to fall out of phase. Also, the uncertainty in pendulum length impacts the natural frequency, and the uncertainty in the logarithmic decrement affects the calculated damping ratio. Small uncertainties in these quantities can become more visible over multiple oscillation cycles.

The methods used to find damping introduces another limitation. The damping constants were obtained from decay measurements of the uncoupled system and then applied to the coupled simulation. This approach assumes that the damping behavior does not change once the two components are connected together. Furthermore, the calculation of the logarithmic decrement from the experimental peaks also depends on accurate identification of the peaks. Extra measurement noise and taking measurements from a limited number of peaks impact the calculated damping coefficients and the rate of decay of the oscillations.

Several improvements for the methods could bring the physical experiment and simulations closer together. Instead of arbitrary values, the Simulink model should use the experimental data for the cart position and pendulum angle. Using this data would ensure that the resulting simulation responses are closer to the actual experimental model. The spring constant could also be measured differently. Instead of a single or double force input, numerous inputs and linear regression should be used to reduce measurement error. To improve accuracy, the damping parameters could also be calculated using a larger number of oscillation peaks. Friction in the cart and pendulum contact locations should also be more accounted for. This could be done using a damping model that includes the friction as the experimental data shows that viscous damping alone is insufficient.

C. Experimental vs Simulated Responses

Comparison between the nonlinear experimental response and the Simulink linearized model provided a measure of how accurately the derived equations represented the physical system. As shown in Fig. 8 below and Fig. 5, both responses showed a decaying oscillator motion with a decreasing amplitude towards the equilibrium point.

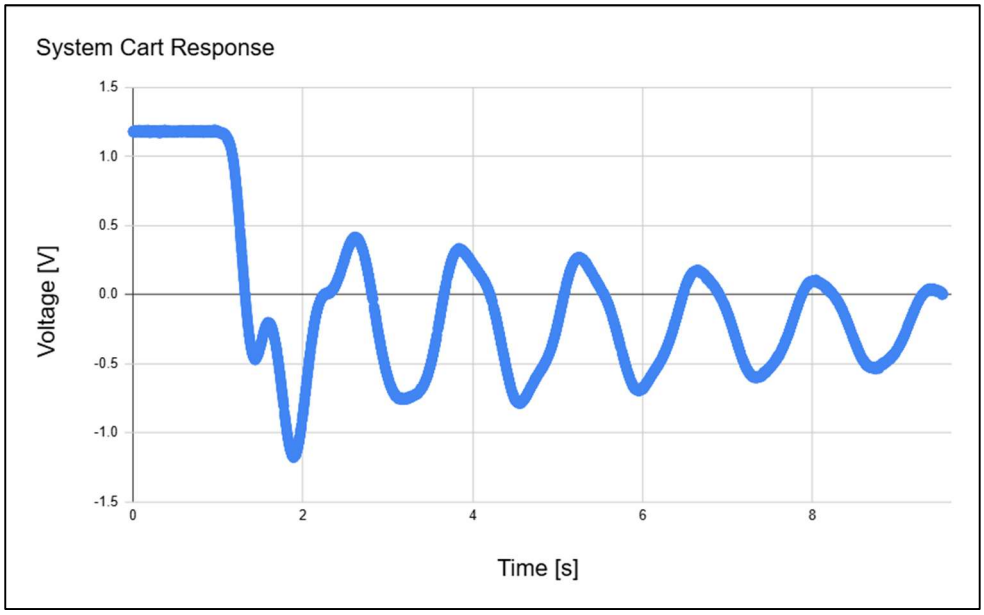


Fig. 8. Isolated pendulum response

The experimental oscillation period was approximately 1.4 seconds when compared to the linearized Simulink period of about 2 seconds. The Simulink model also predicted a slower decay when compared to the experimental measured response. Differences in the rate of decay might suggest that the damping coefficients were not modeled as accurately.

However, differences in timing and magnitude of the peaks were expected as the mathematical model is an idealized representation of the real life pendulum system. The model represents the cart damping and the pendulum damping using the terms, $-c\dot{x}$ and $b\dot{\theta}$, respectively. In the experimental system, there are other factors not considered by the Simulink. For example, the cart and rail friction is not perfect, there could be small alignment errors, and/or other effects that are not directly proportional to the velocity of the cart. The pendulum's physical rotational contact may also have friction that is not easily representable by the single damping coefficient provided to the model. These factors can directly alter the decay rate and response, even if the governing equations represent the dynamics correctly.

The linear and nonlinear Simulink simulations provided an additional way to evaluate the nonlinear terms in the governing equations. Linearization assumes that the pendulum angle remains small and that higher order terms can be neglected. Under small oscillations the linear and nonlinear simulations should produce similar responses. As the pendulum approached equilibrium, the simulations from the two models became more similar. This behavior is consistent with small angle approximation used as the linearized expressions become more accurate as the system approaches zero (smaller oscillations).

D. Above and Beyond, Phase Diagrams

The analysis of motion for a dynamic system can be simplified by utilizing a visualization known as a Phase Diagram. These plots put a system's characteristic motion against its velocity, allowing one to quickly understand the transfer between kinetic and potential energy. This is relevant to the system which was constructed for this experiment as the angle of the pendulum can serve as the positional marker. While the angular velocity was not recorded during the experiment it can be calculated by taking a Riemann approximation of the angle data with the frequency of data taken.

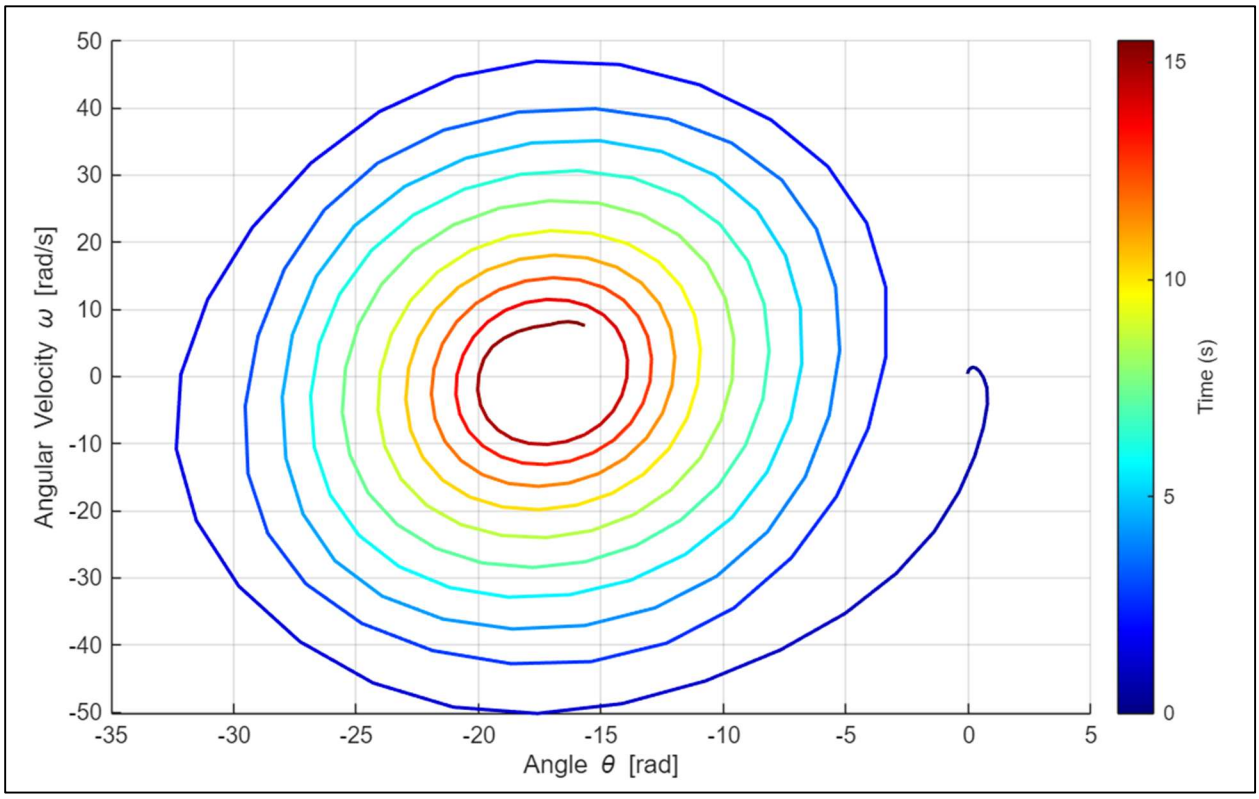


Fig. 9. Phase Diagram of Pendulum Motion

The above diagram displays the motion of the pendulum as it is allowed to return to a point of rest. Note that the cart was locked in place and not allowed to move. As the system loses energy and transfers it to its surroundings per the second law of thermodynamics, it decays and forms the spiral shape. The cylindrical nature is representative of the interchange of angle and angular velocity or more simply potential and kinetic energy. The pendulum being at the apex of its swing (when it is about to pivot into the other direction) is represented by the locations of zero velocity. When the pendulum is at the trough of its swing and at its highest speed the angle is about

-17.5 degrees. Note that the angle is not zero as the probe measures zero as the first position of the pendulum, which during the experiment was when the pendulum was held to one side.

When the entire system is mapped in a phase diagram the motion is noticeably more complex and unpredictable. This is due to the introduction of the sliding cart serving as another body to transfer energy through. The cart has its own exchange of kinetic energy and potential energy as the motion of the pendulum results in its spring being compressed and uncompressed. This also increases the production of entropy as the motion of the cart generates more friction, and hence more energy is lost to the surroundings. Note how a similar point of rest is reached in about 10 seconds for this trial when in the last it took 15 seconds.

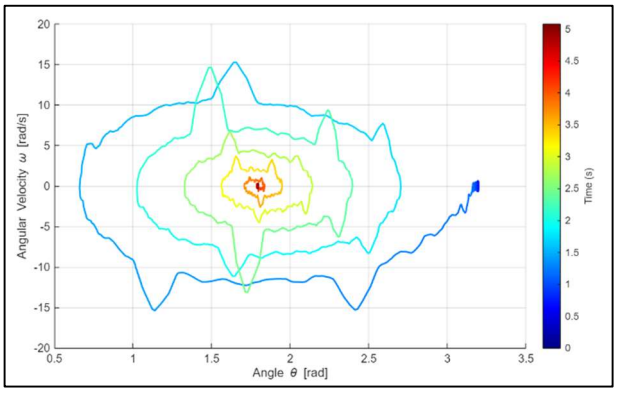


Fig. 10. Phase Diagram of cart motion without additional mass

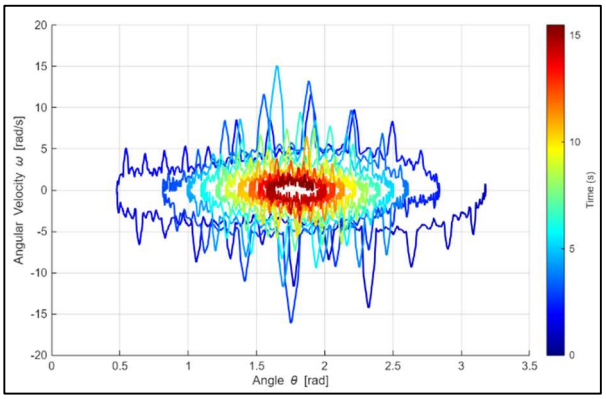


Fig. 11. Phase diagram of cart motion with additional mass.

Lastly, phase diagram analysis of just the motion of the cart clues in some of the chaos of the phase diagram of the entire system. Two cart cases were analyzed, one with extra mass added to the pendulum (left) and one without. The test was conducted by pulling the cart all the way to one side and letting go without touching the pendulum. As is visible in the plot the pendulum mass begins swinging due to the motion of the cart. When the pendulum and cart swing in the same direction it creates constructive interference of acceleration, resulting in those large spike of velocity in the phase diagram. When the pendulum has extra mass it introduces extra potential energy into the system at the starting position. This is clear in the diagrams as it takes the extra mass system about 15 seconds to reach a relative point of rest and the original case only 5 seconds. It takes the same dampening longer as it has to transfer more energy out of the system.

VI. Conclusion

The pendulum cart coupled system is a deceptively complex system. While being complex, it holds concepts that apply everywhere. Deriving equations of motion, and being able to identify important system parameters, allows for further understanding of dynamic systems. These systems are crucial for everyday applications such as autopilot, or automatic traction control. The development and improvements of simulation software still require the equations of motion to work. Further expansion on this topic could be applying simulation data back into the system to alter its behavior.

Appendix

A. Derivation of the non-linear equations of motion

Pendulum Mass Motion

Horizontal

$$x_p = x + R\sin\theta$$

$$\dot{x_p} = \dot{x} + R\ddot{\theta}\cos\theta$$

Vertical

$$y_p = -R\cos\theta$$

$$\dot{y_p} = R\dot{\theta}\sin\theta$$

Find Velocity

$$v_p^2 = x_p^2 + {y_P}^2$$

$$v_p^2 = (\dot{x} + R\ddot{\theta}\cos\theta)^2 + (R\dot{\theta}\sin\theta)^2$$

Expand, reduce using

$$cos^2\theta + sin^2\theta = 1$$

$$v_p^2 = \dot{x}^2 + 2R\dot{x}\dot{\theta}cos\theta + R^2\dot{\theta}^2$$

Kinetic energy T

$$T_1 = \frac{1}{2}M_1\dot{x}^2$$

$$T_2 = \frac{1}{2}M_2 v_p^2$$

$$T_{total} = \frac{1}{2}(M_1 + M_2)\dot{x}^2 + M_2R\dot{x}\dot{\theta}cos\theta + \frac{1}{2}M_2R^2\dot{\theta}^2$$

Potential Energy

$$V = \frac{1}{2}kx^2 - M_2gRcos\theta$$

Construct the Lagrange

$$L = T - V$$

$$L = \frac{1}{2}(M_1 + M_2)\dot{x}^2 + M_2R\dot{x}\dot{\theta}cos\theta + \frac{1}{2}M_2R^2\dot{\theta}^2 - \frac{1}{2}kx^2 - M_2gRcos\theta$$

$$Q_x = -c\dot{x} \;\; and \;\; Q_\theta = -b\dot{\theta}$$

Equation for cart x

$$\frac{\partial L}{\partial \dot{x}} = (M_1 + M_2)x + M_2\dot{R}\dot{\theta}cos\theta$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = (M_1 + M_2)\ddot{x} + M_2R\theta cos\ddot{\theta} - M_2R\dot{\theta}^2 sin\theta$$

Substitute spring equation and rearrange based off Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = -c\dot{x}$$

$$(M_1 + M_2)\ddot{x} = M_2R\dot{\theta}^2 sin\theta - M_2R\ddot{\theta}cos\theta - c\dot{x} - kx$$

Equation for Pendulum θ

$$q = \theta$$

$$\frac{\partial L}{\partial \dot{\theta}} = M_2Rxcos\theta + \dot{M}_2R^2\dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = M_2R\ddot{x}\cos\theta - M_2R\dot{x}\dot{\theta}\sin\theta + M_2R^2\ddot{\theta}$$

Differentiate the terms with respect to θ

$$\frac{\partial L}{\partial \theta} = -M_2R\dot{x}\dot{\theta}\sin\theta - M_2gR\sin\theta$$

Using the following Lagrange formula, cancel, rearrange, divide by R:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = -b\dot{\theta}$$

$$M_2R\ddot{\theta} = -M_2\ddot{x}\cos\theta - \frac{b}{R}\dot{\theta} - M_2g\sin\theta$$

B. Linearization of non-linear equations of motion

$$(M_1 + M_2)\ddot{x} + c\dot{x} + kx = M_2R\dot{\theta}^2 sin\theta - M_2R\ddot{\theta}cos\theta$$

$$M_2R\ddot{\theta} = -M_2\ddot{x}cos\theta - \frac{b}{R} - M_2gsin\theta$$

Small angle approximation derivation

$$\sin(\theta) \approx \sin(\theta) - \cos(\theta)\,(\theta - 0)$$

$$\sin(\theta) \approx \sin(0) - \cos(0)\,(\theta - 0)$$

$$\sin(\theta) \approx \theta$$

$$\cos(\theta) \approx \cos(\theta) - \sin(\theta)\,(\theta - 0)$$

$$\cos(\theta) \approx \cos(0) - \sin(0)\,(\theta - 0)$$

$$\cos(\theta) \approx 1$$

$$\sin(\theta) \approx \theta$$

$$\cos(\theta) \approx 1$$

Final linearized equations of motion

$$(M_1 + M_2)\ddot{x} + c\dot{x} + kx = M_2 R\dot{\theta}^2\theta - M_2 R\ddot{\theta}$$

$$M_2 R\ddot{\theta} = -M_2\ddot{x} - \frac{b}{R} - M_2 g\theta$$

C. Uncertainty experimental values

Table A1 Uncertainty calculation values.

Measurement	Average Value	Resolution	Units
Mass Weight (M2)	1.330	0.001	Kilogram
Force Gauge	3.5	0.2	Newton
Tape Measure	0.370	0.001	Meter
Ruler	0.0325	0.00079	Meter
Angular Encoder (X1)	23.906238	0.001	Radian
Angular Encoder (X2)	23.378894	0.001	Radian
Distance (X1)	2.703194	0.00079	Meter
Distance (X2)	2.398450	0.00079	Meter

D. Uncertainty Equations

$$\Delta\delta_{Cart} = \delta\sqrt{(\frac{\Delta X_1}{X_1})^2 + (\frac{\Delta X_2}{X_2})^2}$$

$$\Delta\delta_{Pendulum} = \delta\sqrt{(\frac{\Delta Y_1}{Y_1})^2 + (\frac{\Delta Y_2}{Y_2})^2}$$

$$\Delta\zeta_{Cart} = \delta_{Cart}\frac{4\pi^2}{(4\pi^2+\delta_{Cart}^2)^{\frac{3}{2}}}$$

$$\Delta\zeta_{Pendulum} = \delta_{Pendulum}\frac{4\pi^2}{(4\pi^2+\delta_{Pendulum}^2)^{\frac{3}{2}}}$$

$$\omega_{n,Cart} = -\Delta\zeta_{Cart}\frac{\zeta_{Cart}}{(1-\zeta_{Cart}^2)^{\frac{3}{2}}}$$

$$\omega_{n,Pendulum} = -\Delta\zeta_{Pendulum}\frac{\zeta_{Pendulum}}{(1-\zeta_{Pendulum}^2)^{\frac{3}{2}}}$$

$$\Delta k = k\sqrt{(\frac{\Delta F}{F})^2+(\frac{\Delta X}{X})^2}$$

$$\Delta C = \sqrt{(2\Delta\zeta\sqrt{km_1})^2+(\frac{m_1\zeta\Delta k}{\sqrt{km_1}})^2+(\frac{k\zeta\Delta m_1}{\sqrt{km_1}})^2}$$

$$\Delta b = b\sqrt{(2\omega_n m_2R^2)^2+(2\zeta m_2R^2\Delta\omega_n)^2+(2\zeta\omega_n R^2\Delta m_2)^2+(4\zeta\omega_n m_2R\Delta R)^2}$$

E. Uncertainty propagation diagrams

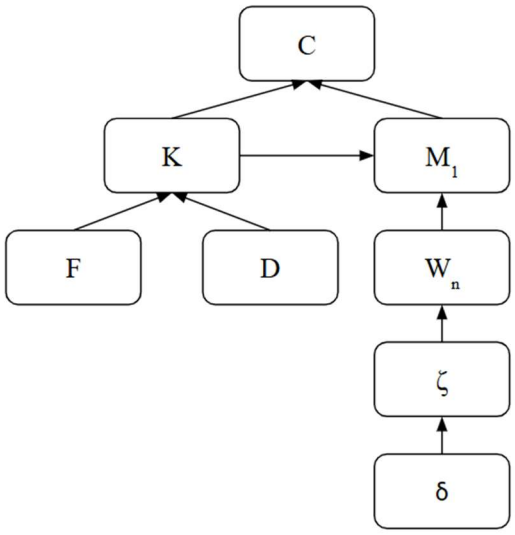


Fig. A1 Uncertainty propagation for damping coefficient c

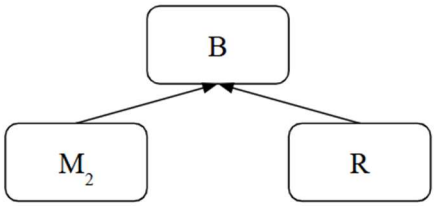


Fig. A2 Uncertainty diagram for damping coefficient b.

References

[1] Wagner, J. R., ME 4440 - Mechanical Engineering Lab III Student Manual, Fall 2026, Department of Mechanical Engineering, Clemson University, Clemson, SC, 2026, pp. 9-12.